

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. (EE) Examination

November 2013

EE404 Energy Management and Conservation

Date: 19.11.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION - I

Q - 1 (1) Draw Classification chart of furnaces and Explain performance evaluation of typical furnace. [07]

Q - 2 (a) Explain Need for Electrical Load Management and Step By Step Approach for Maximum Demand Control. [07]

(b) Explain the seven principles of energy management and Draw flow chart of energy planning. [07]

## OR

Q - 2 Write Short notes: (Any Two) [14]

- (1) Energy efficient motors
- (2) Energy scenario of India and laws of thermodynamics.
- (3) Energy accounting and analysis.

Q - 3 (a) Draw link or path of transport of energy and Explain energy cost of food, fuel, goods & services. [07]

(b) What is energy management cell? Explain its functions, role and responsibilities. [07]

## OR

Q - 3 (a) Draw the single line diagram of electrical distribution system and Explain compressed Air storage system. [07]

(b) Explain ways to improve energy efficiency in buildings. [07]

Q - 4 (a) Write the points of electrical system optimization and Explain methods to improve plant power factor, conduction loss and switching loss. [07]

Q - 5 (a) Write the Potential Energy Conservation Opportunities in HVAC System. [07]

(b) What are the types of energy audit? Explain them in details with comparison. [07]

OR

Q - 5 (a) Write the Potential Energy Conservation Opportunities in lighting System. [07]

(b) Explain energy monitoring and energy consciousness. [02]

(c) Explain energy Audit of electrical system. [05]

Q - 6 (a) Write the principle of cogeneration and Explain classification of cogeneration system. [07]

(b) Explain overhead and underground pumped hydroelectric energy storage plant. [07]

OR

Q - 6 (a) Explain thermal energy storage. [07]

(b) Explain energy conservation in cement company and electricity generating station. [07]

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